

Pre-Board Examination - 2079

Class : XII

Subject: Mathematics (0081)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet.

[11 × 1 = 11]

- The term independent of x in the expansion of $\left(x + \frac{1}{x^2}\right)^{12}$ is
 - 1st
 - 4th
 - 5th
 - 6th
- The value of $\frac{(\cos 50^\circ + i \sin 50^\circ)(\sin 60^\circ + i \cos 60^\circ)}{(\cos 10^\circ + i \sin 10^\circ)^5}$ is
 - $\frac{\sqrt{3}}{2} + \frac{i}{2}$
 - $\frac{1}{2} + i \frac{\sqrt{3}}{2}$
 - $\frac{\sqrt{3}}{2} - \frac{i}{2}$
 - $\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$
- If $\sin^2 x = \sin^2 \alpha$, then the general solution of x is
 - $n\pi \pm \alpha, n \in \mathbb{R}$
 - $n\pi + (-1)^n \alpha, n \in \mathbb{Z}$
 - $n\pi \pm \alpha, n \in \mathbb{Z}$
 - $2n\pi \pm \alpha, n \in \mathbb{Z}$
- The value of $\tan^{-1} \frac{5}{12}$ is
 - $\sin^{-1} \frac{12}{13}$
 - $\cos^{-1} \frac{12}{13}$
 - $\sec^{-1} \frac{12}{13}$
 - $\operatorname{cosec}^{-1} \frac{12}{13}$

5. A unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ is
 a) $\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$ b) $\frac{\vec{a} \times \vec{b}}{|\vec{a}| \times |\vec{b}|}$ c) $\vec{a} \times \vec{b}$ d) $\frac{\vec{a} \times \vec{b}}{\vec{a} \cdot \vec{b}}$
6. The probability of occurrence of 5 Saturdays in the month of April is
 a) $\frac{1}{7}$ b) $\frac{2}{7}$ c) $\frac{3}{7}$ d) $\frac{5}{7}$
7. The eccentricity of a conic is $\frac{5}{2}$, then the conic must be
 a) a hyperbola b) an ellipse c) a circle d) a parabola
8. $\int \ln e^{\cot x} dx$ is equal to
 a) $e^{\cot x} + C$ b) $\ln(\sin x) + C$ c) $-\ln(\sin x) + C$ d) $-\operatorname{cosec}^2 x + C$
9. The differential equation $\left[y + \frac{dy}{dx}\right]^{\frac{1}{2}} = \left(\frac{d^2y}{dx^2}\right)^{\frac{2}{3}}$ is of
 a) 2nd order, 4th degree b) 1st order, 4th degree
 c) 2nd order, 3rd degree d) 2nd order, 2nd degree
10. At the end of forward elimination process using Gauss elimination method for solving a system of three linear equations, if the augmented matrix reduces the form $\begin{bmatrix} a_1 & b_1 & c_1 : d_1 \\ 0 & b_2 & c_2 : d_2 \\ 0 & 0 & 0 : d_3 \end{bmatrix}$, $d_3 \neq 0$, then the system has
 a) no solution b) unique solution
 c) two solutions d) infinitely many solutions
11. P and Q are two unlike parallel forces such that Q is one third of P. What is their resultant?
 a) 2Q b) 2P c) $\frac{3P-Q}{3}$ d) $\frac{P-3Q}{3}$

OR

The solution of the difference equation $y_t = 3y_{t-1}$ is

- a) $y_t = 2 \times 3^t$ b) $y_t = 3 \times 3^t$ c) $y_t = 4 \times 3^t$ d) all of these

Group 'B'

Short Answer Questions.

[8 × 5 = 40]

12. a) In how many ways can five couples sit in a row? How many ways can it be done if every husband and his wife sit together? [1+2]

- b) Solve by using row-equivalent matrix method
 $x + y = 5, 2x + 3y = 12$ [2]
13. a) If the roots of the equation
 $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal,
 then show that: $\frac{a}{b} = \frac{c}{d}$. [3]
- b) If G is the set of cube roots of unity. Construct a Cayley's table for G under multiplication. Mention the inverse of each element of G . [2]
14. a) Find the eccentricity, foci and length of latus rectum of the
 ellipse $\frac{(x+2)^2}{16} + \frac{(y-5)^2}{9} = 1$. [1+1+1]
- b) Using vector method find the area of triangle with vertices $A(1, 4, 6)$,
 $B(-2, 5, 1)$ and $C(1, -1, 1)$. [2]
15. a) The equations of two regression lines are $3x + 4y = 65$
 and $3x + y = 32$. Find
 i) the arithmetic means of x and y .
 ii) the regression coefficients.
 iii) the correlations coefficient between x and y . [1+1+1]
- b) What is the probability that a student may fail in exactly one subject
 in an examination of 6 subjects in annual NEB examination? [2]
16. a) Find from first principle, the derivative of $e^{\sin x}$. [3]
- b) Evaluate : $\int \frac{dx}{e^x + e^{-x}}$ [2]
17. a) Evaluate : $\int \frac{dx}{(x+1)(x+2)}$ [2]
- b) Using L'Hospital's rule, evaluate: $\lim_{x \rightarrow 0} \frac{x - \sin x \cdot \cos x}{x^3}$ [3]
18. a) Solve the following equations using Gauss-Elimination Method
 $5x - 3y = 19, 2x + 5y = -11$ [2]
- b) Using simplex method, maximize $Z = 5x - 3y$ subject to
 $3x + 2y \leq 6, -x + 3y \geq -4, x, y \geq 0$ [3]

19. Find the velocity and direction of projection of a shot which passes in a horizontal direction just over the top of a wall which is 250 m off and 125 m high. ($g = 9.8 \text{ ms}^{-2}$) [5]

OR

The demand equation of a certain commodity is $P = 45 - 2Q - Q^2$, where P is the price and Q is the quantity. Find consumer's surplus when

- (i) demand is 3 (ii) the price is 30. [2+3]

Group 'C'

Long Answer Questions.

[3 × 8 = 24]

20. a) Show that:
$$\sum_{n=1}^{\infty} \frac{n(n+1)}{n!} = 3e$$
 [3]

- b) For a series $1.2 + 2.3 + 3.4 + \dots$

i) find the n^{th} term

ii) find the sum of first n terms

iii) Using principle of mathematical induction, prove that the sum of

first n terms of the given series is $\frac{n(n+1)(n+2)}{6}$ [1+1+3]

21. a) If $\cos^{-1} x + \cos^{-1} y = \theta$, show that $x^2 - 2xy \cos \theta + y^2 = \sin^2 \theta$. [3]

- b) Mention the condition for the two planes to be perpendicular to each other. Find the equation of plane through the points $(-1, 1, 1)$ and $(1, -1, 1)$ and perpendicular to the plane $x + 2y + 2z = 5$. [1+4]

22. a) Examine whether the function $f(x) = x^2 - 6x + 1$ satisfies Lagrange's Mean Value Theorem [MVT] on $[1, 3]$. If it satisfies MVT, find the coordinates of the point at which the tangent is parallel to the chord joining the points $A(1, -4)$ and $B(3, -8)$. [4]

- b) Identify whether each of the given differential equation is linear **or** homogeneous? Solve any one of them:

i) $\frac{dy}{dx} = \frac{y}{x} - \sin^2 \frac{y}{x}$ ii) $\tan x \frac{dy}{dx} + y = \sec x$ [1+3]

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